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Inherent impact of FDM additive manufacturing processes on the magnetic performances of MnZn ferrite☆

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ABSTRACT

The layer-wise deposition intrinsic to extrusion-based additive manufacturing induces mesostructural features that behave as distributed air gaps and can modify the effective magnetic response of printed cores, an aspect that remains insufficiently documented for soft ferrites. In this work, the inherent impact of Fused Deposition Modeling (FDM) printing paths on the magnetic properties of additively manufactured MnZn ferrite cores is investigated experimentally. Toroidal cores were fabricated using a PIM-like FDM process from a polymer-based pellet feedstock loaded with 50 vol% MnZn ferrite powder. Two deposition routes were compared: concentric and rectilinear patterns. Magnetic characterization reveals a clear influence of the deposition path on the effective magnetic permeability. While moderate differences are observed under low-signal excitation, significant deviations appear at higher excitation levels. These variations are attributed to the oriented inter-bead porosities induced by the FDM process, which behave as distributed air gaps within the magnetic circuit. Additional investigations based on under-extrusion highlight a marked difference in sensitivity between the two deposition strategies, with concentric patterns exhibiting a higher robustness to local variations in material deposition. In contrast, magnetic losses remain only slightly affected by the deposition strategy. These results highlight the deposition path as an additional design parameter for tailoring the magnetic behavior of additively manufactured ferrite components.

1. Introduction

In magnetic components subjected to high excitation levels, performance limitations increasingly arise from non-uniform magnetic flux distributions within the core. Localized flux concentration leads to premature saturation, increased magnetic losses and thermal constraints that directly limit achievable power density. Classical solutions based on discrete air gaps efficiently increase energy storage capability, but at the cost of strong fringing fields and additional copper losses [1]. To overcome these drawbacks, several studies have proposed to distribute magnetic reluctance within the core volume rather than concentrating it at a single location. In this perspective, the concept of spatially engineered magnetic permeability has emerged as an efficient alternative, enabling smoother flux distributions and improved thermal behavior. Notably, Byerly et al. demonstrated that strain-induced permeability gradients in soft magnetic metallic ribbon cores allow the redistribution of magnetic flux and the mitigation of thermal hot spots without relying on discrete air gaps [2,3]. Similar objectives have been pursued through multi-permeability magnetic cores combining materials with contrasted permeabilities, which can be interpreted as volumetric implementations of distributed air-gap concepts [4,5].

Despite their effectiveness, existing permeability-engineering strategies remain strongly constrained by the nature of the materials and processing routes involved. Approaches based on mechanical or thermal treatments are essentially limited to metallic magnetic systems, while multi-material core architectures rely on discrete permeability steps and additional interfaces that may restrict design freedom and scalability. In contrast, despite their widespread use in power electronics due to their

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favorable compromise between magnetic permeability and losses [6,7], soft ferrites have rarely been considered as candidates for spatial permeability tailoring. As a result, the potential of introducing controlled, spatially distributed magnetic reluctance within ferrite cores remains largely unexplored, particularly in the context of manufacturing routes capable of generating such features in an intrinsic and reproducible manner.

Additive manufacturing has emerged as a promising alternative to conventional shaping routes for magnetic components, as it offers an unprecedented level of geometric freedom and design flexibility. This design freedom is particularly attractive for the fabrication of non-standard magnetic core geometries, integrated magnetic structures, or components with internal functional features such as cooling channels, which are difficult to obtain through conventional pressing and machining routes [8]. It also enables the design of internal magnetic architectures, such as pseudo-laminated patterns based on space-filling curves, including Hilbert-type motifs [9,10]. Various additive manufacturing techniques have been investigated for magnetic materials. Powder-bed-based processes, such as selective laser melting or sintering, have been widely explored for metallic soft magnetic materials, enabling the fabrication of dense magnetic parts with complex geometries but often at the expense of increased losses due to process-induced microstructural features [9,11]. Vat photopolymerization techniques, including Stereolithography (SLA) and Digital Light Processing (DLP), have also been investigated for ferrite ceramics, offering high shape accuracy and fine feature resolution through the use of highly loaded photo-curable suspensions followed by debinding and sintering steps [12,13]. However, these approaches rely on photo-curable binder systems in which polymer chains are chemically cross-linked during UV exposure. Such cross-linked networks are intrinsically more difficult to remove than thermoplastic binders, as they exhibit limited solubility and require high-temperature thermal degradation. This significantly increases the sensitivity of the debinding stage, often necessitating complex and tightly controlled thermal cycles to prevent cracking or shape distortion, which may restrict process robustness and scalability [14]. In contrast, extrusion-based FDM processes rely on thermoplastic binder systems whose physical nature allows for more straightforward and controllable debinding mechanisms. These processes have attracted growing interest for both metallic and ceramic magnetic materials due to their simplicity, versatility, and relatively low cost [15]. When combined with feedstocks derived from Powder Injection Molding (PIM), FDM enables the shaping of ferrite components through a debinding and sintering route, resulting in fully functional ceramic magnetic parts [16,17].

Beyond purely geometric considerations, additive manufacturing also offers new opportunities for tailoring the magnetic behavior of components through controlled mesostructural features. The layer-by-layer deposition inherent to FDM-based processes intrinsically induces a specific mesostructure resulting from the discrete placement of individual filaments [18]. Unlike conventional shaping routes, where the material can be considered macroscopically homogeneous after sintering, the deposition of adjacent tracks leads to the formation of inter-bead porosities whose size, distribution, and orientation are governed by the printing strategy [19]. These mesostructural features are an inherent consequence of the process itself.

In ferrite components manufactured through PIM-like additive routes, these inter-bead porosities are only partially eliminated during debinding and sintering and therefore persist in the final ceramic parts [20]. Their spatial organization reflects the chosen deposition path, leading to anisotropic mesostructural patterns at the scale of the deposited filaments. As a result, the internal architecture of the printed core cannot be considered isotropic, even when the global density remains comparable to that obtained by conventional processes. This raises the question of whether the deposition path itself may act as an intrinsic processing parameter controlling the magnetic behavior of additively manufactured ferrite components.

The influence of additive manufacturing parameters on the magnetic properties of printed components has been investigated in several studies, primarily focusing on metallic magnetic materials produced by extrusion-based processes [16,21]. In these works, the printing path and build orientation were shown to induce anisotropic magnetic behavior. Such results highlight that, beyond material composition, the manufacturing strategy itself can significantly affect the effective magnetic response. In contrast, similar investigations remain scarce for soft ferrites, and particularly for MnZn ferrites manufactured through FDM.

The objective of the present work is to experimentally investigate the influence of the deposition path inherent to FDM additive manufacturing on the magnetic behavior of MnZn ferrite cores. To specifically examine the contribution of this effect, toroidal samples were manufactured using two distinct printing strategies (concentric and rectilinear patterns) while maintaining identical manufacturing conditions. The magnetic behavior of the fabricated cores is characterized under both low- and high-signal excitation conditions, allowing the respective influence of the deposition path to be assessed in different magnetic regimes. Particular attention is paid to the relationship between the mesostructural features induced by the printing strategy and the resulting effective magnetic properties. In addition, the sensitivity of the magnetic response to local variations in material deposition is examined through a comparative analysis of slightly under-extruded samples, in order to assess how deviations in filament deposition affect the magnetic performance depending on the printing path.

The article is organized as follows. Section 2 presents the motivations of the study, together with the experimental protocols used for sample fabrication and magnetic characterizations. Section 3 reports and discusses the experimental results, including mesostructural observations, magnetic measurements, and the impact of under-extrusion on the magnetic behavior for each deposition strategy. Finally, the main conclusions and perspectives of this work are summarized in Section 4.

2. Experimental approach and methodology

2.1. Motivations, hypotheses and scope of the study

FDM-based additive manufacturing inherently induces a network of inter-bead porosities originating from the discrete deposition of molten filaments, as shown in Fig. 1 (left). These voids are clearly observed on cross-sectional optical micrographs of the printed parts shown in Fig. 1 (right), called green parts. In ceramic or ferrite parts produced through FDM routes, these porosities are only partially eliminated during sintering and therefore persist in the final component. From a magnetic standpoint, such voids can be regarded as air gaps embedded within the magnetic circuit.

The present study investigates the hypothesis that these distributed air gaps do not merely reduce the effective permeability in an isotropic manner, but that their orientation relative to the magnetic flux path plays a key role. In particular, depending on the deposition strategy, the inter-bead porosities may be oriented either parallel or transverse to the main magnetic flux, potentially leading to markedly different magnetic responses.

To isolate this effect, the scope of the study was to analyze a simple geometry (toroidal cores) for two deposition strategies: a concentric and a rectilinear pattern. The concentric deposition pattern shown in Fig. 2 (left) follows the toroidal geometry of the core. In contrast, in order to avoid the presence of concentric inner and outer walls that could act as preferential magnetic flux paths, the rectilinear configuration was obtained by printing a plate with a fully rectilinear deposition pattern and subsequently machining it into a toroidal geometry, as illustrated in Fig. 2 (right). Although local machining-induced defects cannot be fully excluded, optical inspection after machining and post-sintering tomographic observations did not reveal macroscopic cracking or severe distortion of the rectilinear cores. All other parameters, including material composition and printing conditions were kept strictly identical,

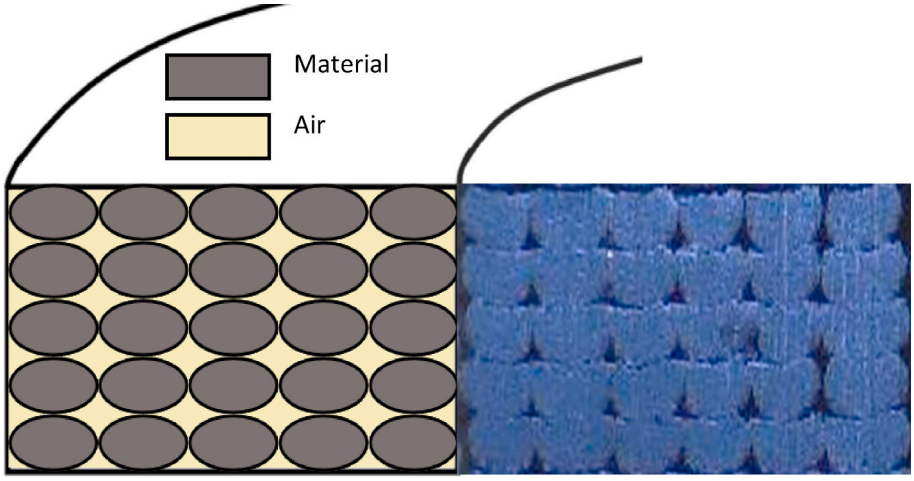


Fig. 1. Schematic representation of inter-bead porosities induced by the FDM-based PIM-like process (left) and cross-sectional optical micrograph of a green MnZn ferrite toroid showing the filament arrangement and associated porosity (right).

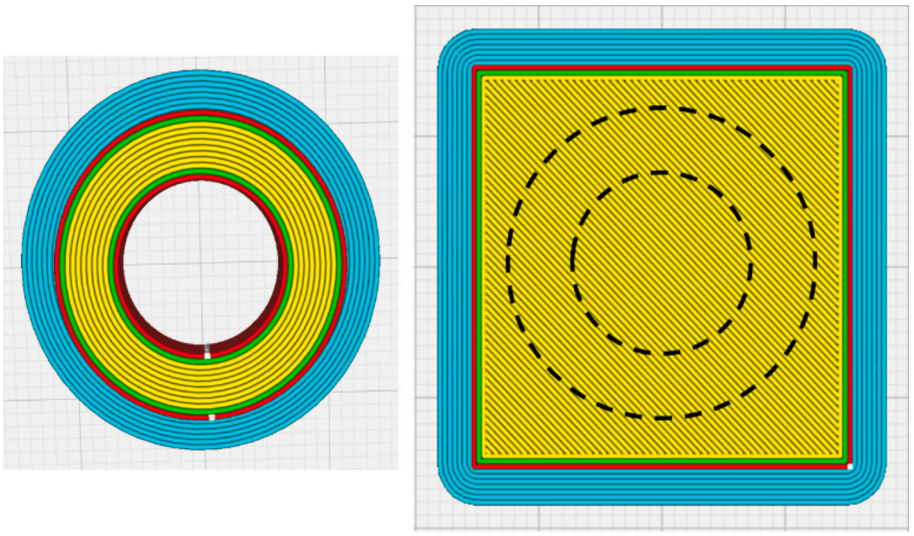


Fig. 2. Concentric (left) and rectilinear (right) printing path strategies used for the fabrication of MnZn ferrite toroidal cores. The two slicings are obtained with Cura Ultimaker slicer.

and the cores were debinded and sintered in the same batch under the same conditions. In addition, for each deposition strategy, one core was printed with a reduced extrusion flow in order to assess the sensitivity of the magnetic behavior to local variations in material deposition while maintaining the same overall printing strategy. Under these conditions, any difference in magnetic behavior can reasonably be attributed to the deposition pattern itself.

2.2. Materials, manufacturing process and magnetic characterization

MnZn ferrite cores were manufactured using a PIM-like FDM process based on the extrusion of a highly filled thermoplastic feedstock. The starting material consisted of a commercial MnZn ferrite powder (Z70L2G, from Japan Metals and Chemicals) initially developed for conventional pressing routes, which was incorporated into a polymeric matrix to enable extrusion-based shaping. The feedstock was subsequently prepared by mixing the ferrite powder with a linear low-density polyethylene (LLDPE)–based binder designed to ensure adequate flowability during extrusion while maintaining a high ceramic loading. The resulting compound, corresponding to a ferrite volume fraction of approximately 50 vol%, was homogenized and processed using a twin-screw extruder to obtain pellets suitable for fused granule fabrication. Further details on the feedstock formulation and processing route can be found in [7].

Additive manufacturing was performed using an FDM printer equipped with a fused granule fabrication print head. A miniaturized pellet extruder developed in-house was mounted on an open-source three-axis manufacturing platform (E3D Tool Changer) [15], enabling the direct extrusion of highly filled thermoplastic feedstocks in granular form. The extrusion nozzle diameter was set to 0.6 mm. The pellet extruder operates using a screw-driven feeding system that conveys the granules toward a heated chamber, where the material is progressively heated to reach the rheological state required for controlled extrusion through the nozzle. The nozzle temperature was set to 190 °C, while the heated build plate was maintained at 80 °C. Toroidal cores were printed using two distinct deposition strategies: a concentric path following the toroidal geometry and a rectilinear path composed of straight deposition lines. For the rectilinear configuration, a flat plate was first printed using a fully rectilinear deposition path and subsequently machined into a toroidal geometry using a CNC milling machine (ShopBot PRS Standard) in order to obtain toroidal cores with dimensions identical to those of the concentric samples. This approach avoids the presence of concentric

inner and outer walls that would otherwise be generated during direct toroidal printing and could act as preferential magnetic flux paths, thereby biasing the evaluation of the rectilinear deposition strategy. All printing parameters were kept strictly identical for both configurations and are summarized in Table 1. For each strategy, three cores were fabricated under identical processing conditions, except for the extrusion flow. Two cores were printed using the reference flow, as determined from preliminary calibration tests to achieve a nominal material deposition flow, while one was printed with an extrusion flow reduced by 15% relative to this reference value.

Fig. 3 presents complementary observations of the green cores obtained with the two deposition strategies. X-ray radiographs were acquired at 100 kV and 81 mA, with a voxel size of 24 μm. The last-layer optical micrographs (top) and the plan-view X-ray radiographs (bottom) consistently reveal distinct track organizations for concentric and rectilinear deposition paths. While their overall volume fraction remains comparable, their spatial distribution and orientation clearly differ depending on the deposition path.

After printing, the green parts underwent debinding and sintering sequences. Debinding enables the removal of the organic binder and the formation of a porous ceramic skeleton [22], whereas sintering allows the grains to coalesce and consolidates the ferrite structure through diffusion-driven densification [23]. Debinding was initiated by a solvent extraction step in n-hexane, aimed at removing the soluble fraction of the binder system and generating an interconnected open-porosity network within the remaining polymer matrix. After drying, thermal debinding was carried out in air (400 °C) to eliminate the residual polymer backbone. The debound parts were then sintered at high temperature (1145 °C, 4 h under 0.7% oxygen) to promote diffusion between adjacent ferrite particles, leading to grain growth and densification of the ceramic structure. The debinding and sintering conditions used in this work were previously validated for MnZn ferrite components and are described in detail in [7]. These thermal conditions were selected based on the PIM processing route previously developed for similar MnZn ferrite components using the same type of commercial powder, in order to ensure compatibility with the powder composition and the binder-removal sequence.

After sintering, the cores were further examined by X-ray computed tomography. Representative top-view slices and corresponding reconstructed cross-sectional views are presented in Fig. 4 for both rectilinear and concentric deposition paths. These qualitative observations show that the overall toroidal geometry is preserved after debinding and sintering, with no visible macroscopic cracking or severe distortion at the investigated scale. Mesostructural heterogeneities remain observable after sintering.

The bulk density and open porosity of the reference-flow sintered cores were determined using an Archimedes method. As summarized in Table 2, concentric and rectilinear cores exhibit very similar density and compaction rates after sintering, with only minor differences in open porosity, in good agreement with the datasheet values. This suggests that the observed differences in magnetic behavior are not primarily governed by global densification effects.

Magnetic characterization was carried out at room temperature under both low and high-signal excitation conditions. Low-signal measurements were performed over a wide frequency range using an impedance analyzer (Agilent 4294 A) with a 500 mV excitation signal and a 9-turn winding. High-signal magnetic measurements were conducted using a dedicated experimental setup designed for the characterization of ferrite cores under dynamic excitation. The bench consisted of a function generator (Metrix GX240) driving a power amplifier (Amplifier Research 75A250A), which supplied the excitation current $i_1(t)$ to the primary winding of the core. The magnetic response was monitored using a digital oscilloscope (LeCroy HRO 66Zi) equipped with calibrated voltage and current probes to simultaneously record the voltage $v_2(t)$ induced in the secondary winding and the current flowing in the primary winding. From these time-domain signals, the magnetic flux density B(t) and the magnetic field strength H(t) were determined in (1) and (2) using the standard relations for toroidal cores, allowing the reconstruction of dynamic B—H hysteresis loops:

$$B(t) = \frac{1}{N_2 S}\int v_2(t)dt \tag{1}$$

$$H(t) = \frac{N_1 i_1(t)}{l_m} \tag{2}$$

where N_1 and N_2 are the number of turns of the primary and secondary windings, S is the cross-sectional area of the core, and l_m is the mean magnetic path length.

The volumetric magnetic losses (P_v) were evaluated from the instantaneous product of voltage and current signals averaged over one excitation period, as explained in (3):

$$P_v = \frac{1}{V}\langle v_2(t) i_1(t)\rangle \tag{3}$$

where V is the volume of the magnetic core.

This setup enables a consistent comparison of the magnetic behavior of the different cores under high-signal excitation conditions.

3. Results and discussion

The results first focus on the mesostructural features induced by the printing strategy and by controlled variations in material deposition, before discussing their impact on magnetic behavior. In addition to comparing concentric and rectilinear deposition paths under nominal printing conditions, the influence of controlled under-extrusion was also investigated. In extrusion-based additive manufacturing, slight deviations in extrusion flow may arise in practice from feedstock inhomogeneities, rheological variations, or instabilities during printing. Introducing an under-extruded core for each deposition strategy therefore provides a relevant means of probing how sensitive the resulting magnetic behavior is to such variations, while keeping all other processing parameters strictly identical. This approach allows evaluating whether the impact of local porosity amplification differs depending on the orientation and organization of the deposited filaments.

Fig. 5 presents X-ray radiographs of the green cores fabricated under reference-flow and under-extruded (by 15%) conditions for both deposition strategies. In both cases, under-extrusion leads to a clear accentuation of inter-bead gaps, which appear as higher-contrast regions associated with reduced X-ray attenuation. Compared with the reference-flow samples, the under-extruded concentric and rectilinear cores exhibit densities of 4.57 g/cm^3 and 4.59 g/cm^3, corresponding to relative decreases of approximately 3.4% and 2.1%, respectively. While this decrease in density remains moderate, the pronounced contrast observed in the radiographs indicates that the missing material is preferentially localized at inter-bead interfaces. As a result, the impact of under-extrusion on magnetic behavior is expected to be governed not only by the reduction in average density, but also by the altered spatial organization of porosity.

The low-signal magnetic response was first examined through complex permeability measurements. Fig. 6 presents the frequency

Table 1
Printing parameters of the cores manufacturing.

Printing parameter	Value
Nozzle Ø [mm]	0.6
Line width [mm]	0.64
Nozzle temperature [°C]	190
Plate temperature [°C]	80
Layers height [mm]	0.2
Printing speed [mm/s]	15

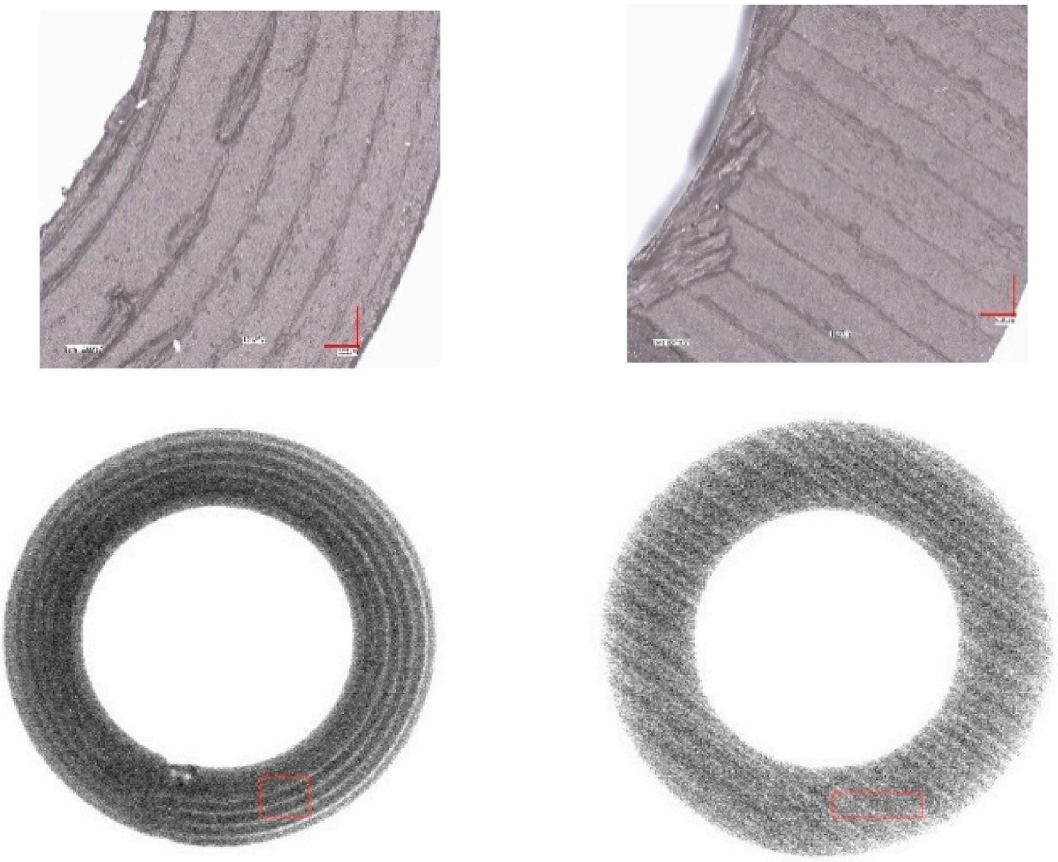


Fig. 3. Morphological observations of green MnZn ferrite toroids printed with concentric (left) and rectilinear (right) deposition paths: last-layer optical micrographs (top) and plan-view X-ray radiographs (bottom).

dependence of the real part of the complex permeability μ' measured on MnZn ferrite cores printed with concentric and rectilinear deposition paths, including both reference-flow and under-extruded samples. Under low-signal excitation, the magnetic response of the cores exhibits a moderate dependence on the deposition path. For the reference-flow printed cores, concentric samples display higher values, with a mean permeability of approximately $\mu'_{rf,\ concentric} = 1003$, whereas rectilinear samples show slightly lower values around $\mu'_{rf,\ rectilinear} = 932$. For comparison, the Z70L2G datasheet reports a nominal low-signal permeability of approximately $\mu' = 1400$. The lower values measured on the FDM-processed cores should be interpreted with caution, since the thermal history, densification state and porosity of the printed cores may differ from those of the datasheet reference material. Measurements performed on two samples for each configuration show a good overall reproducibility and a separation between the two deposition strategies over the investigated frequency range, with concentric cores consistently exhibiting higher permeability than rectilinear ones. The relatively small overall differences observed in this low-signal regime can be attributed to the limited magnetic excitation, which results in negligible saturation effects and an insufficient mobilization of these distributed air gaps.

When considering the under-extruded samples, this trend remains unchanged. The under-extruded concentric core exhibits permeability values comparable to those of the reference-flow concentric samples ($\mu'_{ue,\ concentric} = 1024$), indicating that moderate under-extrusion does not significantly affect the magnetic response in this configuration. In contrast, the under-extruded rectilinear core shows a more noticeable decrease in permeability compared with the reference-flow rectilinear samples ($\mu'_{ue,\ rectilinear} = 752$), suggesting a stronger sensitivity to deposition defects. This behavior is consistent with the hypothesis that inter-bead porosities act as distributed air gaps preferentially aligned with the deposition direction. In concentric deposition paths, these gaps are predominantly oriented parallel to the magnetic flux path, thus limiting their impact on flux conduction. Conversely, in rectilinear configurations, the gaps are more frequently oriented transverse to the flux path, leading to a stronger perturbation of the magnetic circuit when their size increases due to under-extrusion.

The magnetic response of the cores was investigated under increasing excitation by measuring the maximum magnetic induction B_{max} as a function of the maximum applied magnetic field H_{max}, as shown in Fig. 7. The results obtained reveal two distinct groups, with concentric cores systematically reaching higher induction levels than rectilinear ones over the whole investigated field range. The mean relative difference between the two configurations is also reported and increases with the applied magnetic field before stabilizing after the knee of saturation.

For the reference-flow samples, at low magnetic field levels, the difference in B_{max} between concentric and rectilinear cores remains limited, in line with the low-signal permeability results. As the magnetic field increases, this gap progressively widens. This behavior can be interpreted as follows: at low magnetic fields, the flux primarily follows the ferrite matrix, and the distributed air gaps act as a small perturbation. As the excitation increases, local saturation effects develop in the ferrite, forcing part of the magnetic flux to cross the air gaps. In this regime, the orientation and connectivity of the inter-bead porosities become critical, leading to a stronger dependence of the effective magnetic response on the deposition path.

The behavior of the under-extruded samples further supports this interpretation. Although the under-extruded concentric core exhibits induction values slightly higher than those of the reference-flow concentric samples, its behavior remains fully contained within the ±20% variability band reported in the powder manufacturer datasheet,

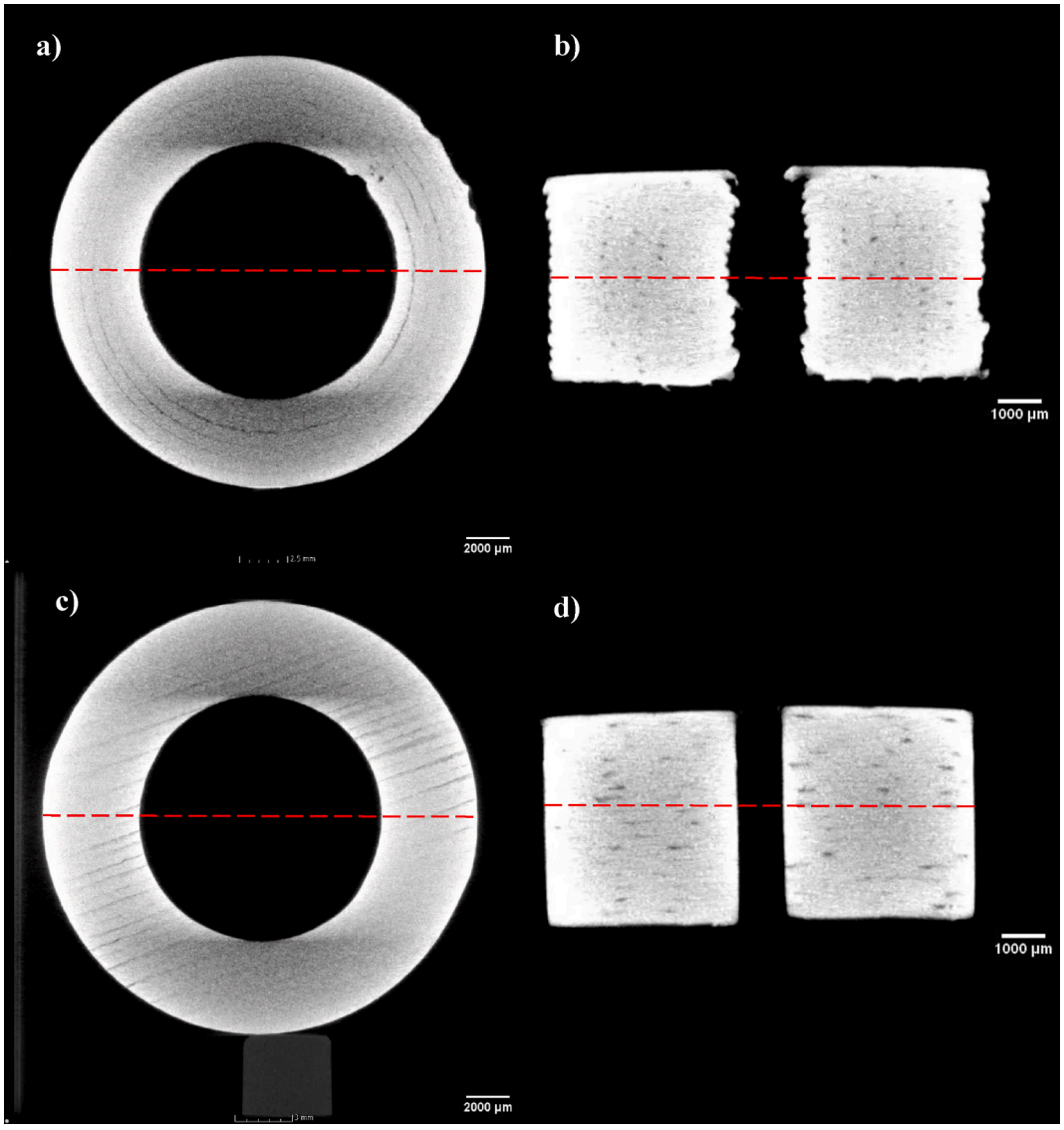


Fig. 4. X-ray computed tomography observations of sintered MnZn ferrite toroidal cores printed with concentric (a, b) and rectilinear (c, d) deposition paths. Top-view slices are shown in (a) and (c), while the corresponding reconstructed cross-sectional views obtained after reslicing are shown in (b) and (d). The dashed lines indicate the position of the reconstructed cross-sections in the top-view slices and the corresponding height of the top-view slices in the cross-sectional views.

Table 2
Density and porosity of the reference-flow MnZn ferrite cores, compared with datasheet values.

Type of core	Mean density [g/cm^3]	Mean relative density rate [%]	Open porosities [%]
Concentrical	4.73	92.7	1.24
Rectilinear	4.69	92.0	2.03
Datasheet	4.70	92.2	–

which defines the expected performance dispersion of the material. The variability bands displayed in Fig. 7 are introduced precisely to account for this intrinsic material dispersion. Therefore, the apparent improvement observed for the under-extruded concentric sample cannot be interpreted as a genuine process-induced performance gain but rather falls within the normal material variability. In contrast, the under-extruded rectilinear core exhibits a systematic reduction of achievable induction compared with reference-flow rectilinear samples, with deviations that extend beyond this ±20% variability range, indicating that the degradation cannot be solely attributed to intrinsic material dispersion. In addition, the rectilinear under-extruded core shows an earlier saturation tendency, visible through a progressive flattening of the B_{max} (H_{max}) curve at high excitation. This confirms that increasing inter-bead gaps primarily penalizes configurations in which porosities are oriented transverse to the magnetic flux path, whereas concentric paths, for which gaps remain mostly parallel to the flux, are less affected by moderate under-extrusion.

The $B_{max}(H_{max})$ curves describe the macroscopic high-field response of the cores and should not be interpreted as a direct measurement of the intrinsic saturation flux density B_s of the ferrite phase. Nevertheless, a comparison was performed under the measurement condition reported in the Z70L2G datasheet, namely H = 1200 A/m, f = 10 kHz and T = 25 °C, as summarized in Table 3. The reference-flow FDM cores reach values of 0.43 T for the concentric path and 0.42 T for the rectilinear path, compared with 0.45 T for the datasheet value. This indicates that, under this specific condition, the induction reached by the FDM cores remains close to the reference value, despite the porosity and the distinct shaping route.

The volumetric magnetic losses were evaluated as a function of the maximum magnetic induction for both deposition strategies, as shown

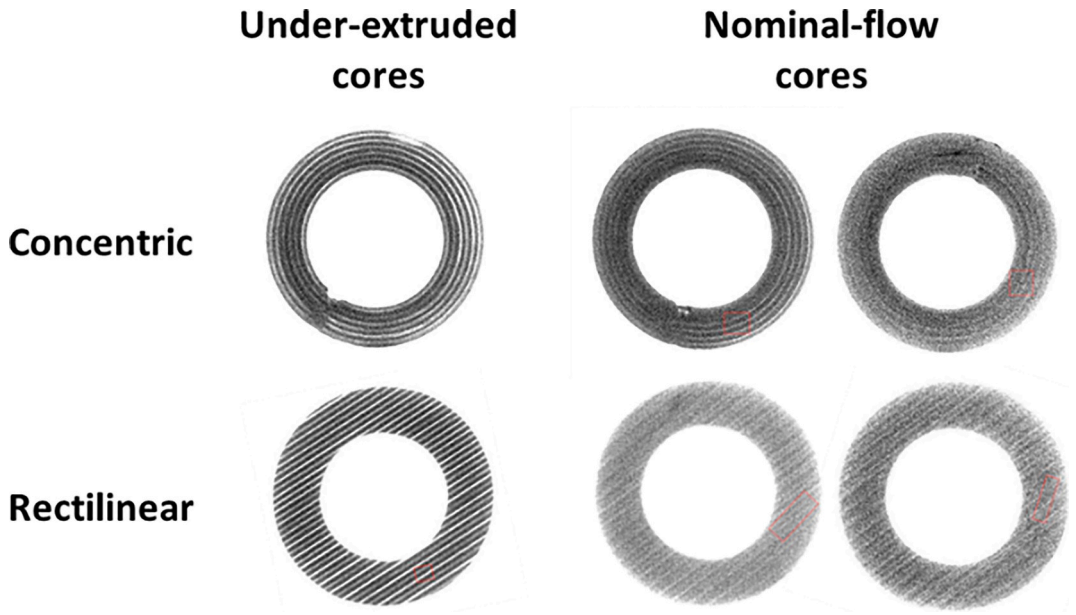


Fig. 5. X-ray radiographs of reference-flow and under-extruded MnZn ferrite cores.

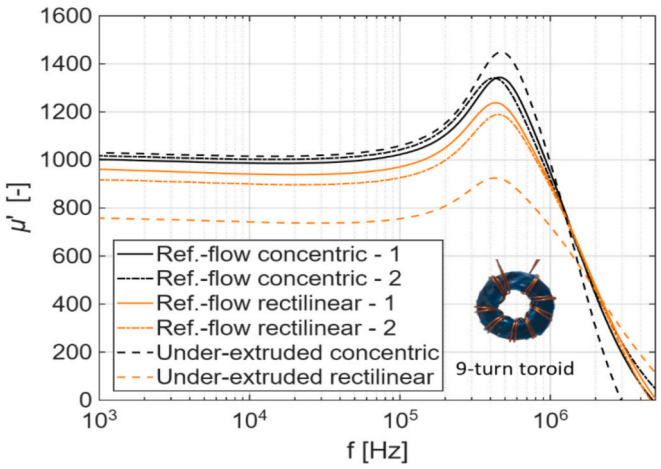


Fig. 6. Real part of the complex permeability μ' as a function of frequency for MnZn cores printed with concentric and rectilinear deposition paths under reference-flow and under-extruded conditions.

in Fig. 8. As an indicative reference, the Z70L2G datasheet reports a loss value of 1100 kW/m^3 at 100 kHz and 200 mT. This value is mentioned only as a reference point, since magnetic losses in MnZn ferrites are strongly dependent on the thermal treatment and on the targeted operating frequency, which is around 500 kHz in this case. As observed for the permeability and high-field magnetic behavior, two distinct groups can be identified for the reference-flow samples. Cores printed with a rectilinear deposition path systematically exhibit higher losses than concentric ones over the investigated induction range. However, the mean relative difference in volumetric losses between the two configurations remains moderate and does not lead to any drastic increase in dissipated power, averaging approximately 16%. This result indicates that, while the deposition path significantly affects the effective magnetic permeability and the high-field response, its impact on magnetic losses is comparatively limited, which is particularly favorable for power electronics applications.

The behavior of the under-extruded samples follows the same general tendency as for previous characterizations. The under-extruded concentric core exhibits losses comparable to those of the reference-flow concentric samples, whereas the under-extruded rectilinear core shows higher losses than the reference-flow rectilinear samples. The difference for under-extruded samples reaches about 51%. This indicates that under-extrusion, through the enlargement of inter-bead gaps, also affects magnetic losses, particularly for rectilinear configurations where these gaps are oriented transverse to the magnetic flux path.

Overall, these results demonstrate that the deposition strategy inherent to FDM-based additive manufacturing directly affects its magnetic behavior through the induced mesostructural anisotropy.

4. Conclusion and perspectives

This work demonstrates that the deposition path intrinsic to FDM-based PIM-like additive manufacturing constitutes an inherent parameter governing the magnetic behavior of MnZn ferrite cores. By isolating the effect of the printing strategy under strictly identical material and processing conditions, clear differences were shown between concentric and rectilinear deposition paths. While the impact on magnetic permeability remains moderate under low-signal excitation, pronounced differences emerge at higher magnetic field levels, where the orientation and connectivity of inter-bead porosities act as distributed air gaps and strongly influence flux propagation. In contrast, magnetic losses are only weakly affected by the deposition strategy within the investigated conditions. The introduction of controlled under-extrusion further reveals a marked difference in sensitivity between the two deposition paths. Whereas the magnetic response of concentric cores remains largely unaffected, rectilinear cores exhibit a significant degradation of magnetic induction under under-extruded conditions.

Overall, this work establishes the deposition path not merely as a technological constraint, but as a design parameter for additively manufactured ferrite components. By controlling the orientation and distribution of mesoscopic porosity through the printing strategy, it becomes possible to tailor the effective magnetic response of ferrite cores, including the deliberate integration of distributed air-gap functionalities. This approach is particularly well suited to the manufacturing of saturable inductors for power electronics converters. The distributed air gap created by additive manufacturing could enable the inductance value to be adjusted according to the current level.

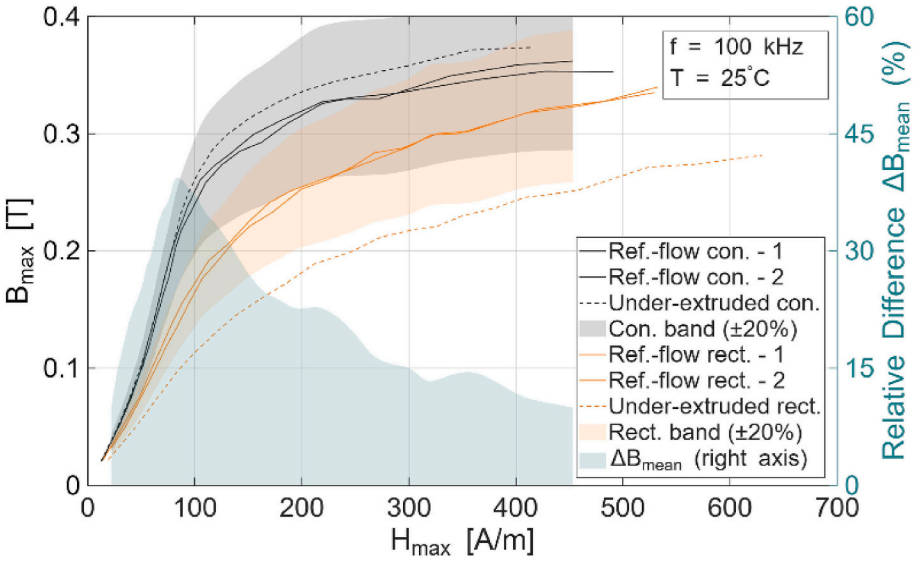


Fig. 7. Maximum magnetic induction B_{max} as a function of maximum magnetic field H_{max} at 100 kHz for MnZn cores printed with concentric and rectilinear deposition paths under reference-flow and under-extruded conditions, including variability bands and mean relative difference between both configurations.

Table 3
Comparison of the magnetic flux density measured under the datasheet condition for the Z70L2G material and for the reference-flow FDM cores.

Sample	Datasheet	Reference-flow concentric	Reference-flow rectilinear
Magnetic flux density [T] @ H = 1200 A/m, f = 10 kHz, T = 25 °C	0.45	0.43	0.42

CRediT authorship contribution statement

Ghali Sqalli: Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Marwane Dherbécourt:** Writing – review & editing, Validation, Methodology, Investigation. **Frédéric Gillon:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Denis Najjar:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Léopold Mikula:** Writing – review & editing, Validation, Investigation. **Vincent Martin:** Writing – review & editing, Methodology, Conceptualization. **Jean-François Witz:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Ulrich Soupremanien:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Abdelkader Benabou:** Writing – review & editing, Supervision, Methodology. **François Lesaffre:** Writing – review & editing, Methodology, Investigation.

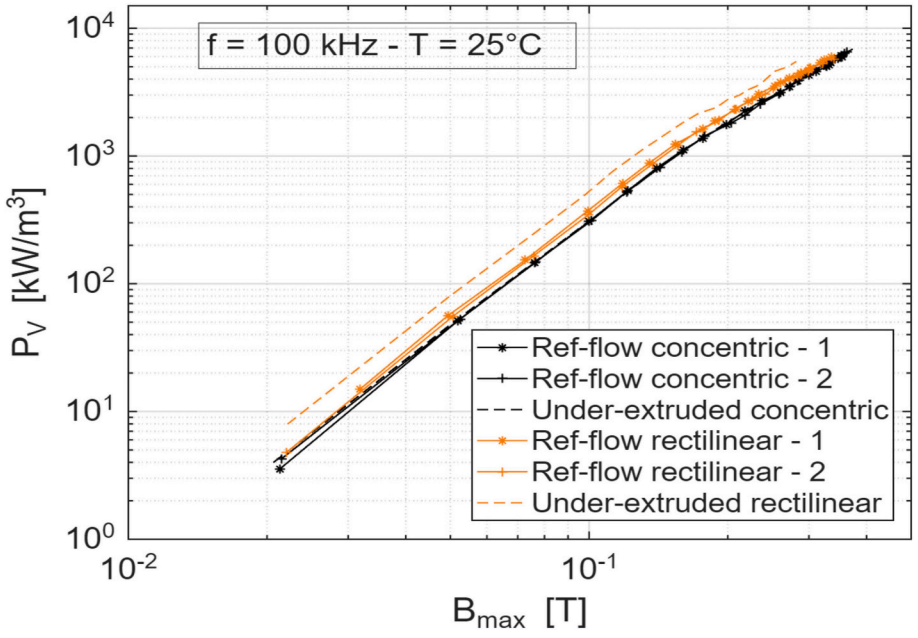


Fig. 8. Volumetric magnetic losses as a function of B_{max} for MnZn ferrite cores printed with concentric and rectilinear deposition paths under reference-flow and under-extruded conditions at 100 kHz.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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